

DNS Walkthrough

DNS stands for domain name system and its job is to translate domain names that are human-readable into IP addresses that are machine-readable. By domain name I mean the human-readable part of a website like `buzzfeed.com` and by IP address I mean Internet Protocol address which is a numerical identifier that allows devices to be reached on a network usually something like `192.0.1.2`

Since the IP address is difficult for humans to remember but a domain name like `buzzfeed.com` is much easier to remember and distribute it is the domain name that is used to locate the IP address that machines use to locate each unique device over the internet. The DNS allows conversion between these 2 systems.

A DNS lookup can involve four types of servers:

1. DNS recursor (resolver) which acts as the librarian. It's job is to receive the request for a certain specific domain and satisfy the request of returning the IP address for it or giving an appropriate error code or timeout message.
2. Root nameserver. It acts like the first locator and carries the directions to the location of the rack of books in our analogous library which has our desired domain/IP address
3. The TLD (top level domain) nameserver. It finds the `.com` or `.org` or `.ai` part of the domain and gives the resolver the authoritative nameserver responsible for that domain.
4. Authoritative nameserver. It acts like a dictionary and houses the definition of all books on the rack. It returns the IP address if it has access to it and is the final major component.

The process between someone typing a website and the host answering is as follows:

User types the hostname/domain name into their browser and this is sent to the DNS recursor -> The recursor librarian sends request to the root server -> the root server sends back the TLD address (`.com` or `.org` etc.) -> the recursor sends request to the `.org/.com` TLD -> the TLD server points to the authoritative nameserver which returns the IP address for the hostname -> The recursor sends back the IP address to the web browser for the requested hostname -> the browser makes request to get the actual webpage -> the browser renders the webpage

DNS caching allows this process to skip steps and allows faster resolution of domain IP addresses. The cache can be housed in places like the user's browser or machine, the internet provider's cache.

A CNAME is when an alias hostname points to another hostname. For example, `blog.medium.com` can point to `medium.com`. Here `medium.com` is the canonical name, and `blog.medium.com` is the alias. The DNS lookup can follow `medium.com`'s records to find the IP address.